LYES TOUMI

AI & Data Engineer | BI Business Analyst



Portfolio

+41 76 522 17 46 • toumilyes15@gmail.com • Rue Charles-Humbert 6, 1205 Geneva • linkedin.com/in/lyes-toumi • github.com/Toumil94 • toumilyes.netlify.app • **Swiss resident** · **B work permit** • EU citizen • Available immediately

PROFILE

5 years of continuous IT and data practice (2021 – present), including **3 years of apprenticeship at EssilorLuxottica** (190,000 employees) on international AI and data projects. **RAG chatbot** designed and deployed to production, cutting information search time by **30%** for the teams. Contributed to a major **BI migration** (Qlik to SAP BusinessObjects) for European business teams. Comfortable across the full cycle: **data engineering** (pipelines, modeling, quality), **BI and decision analytics**, design and deployment of **AI solutions** (LLMs, RAG, agents), and **business analysis** (requirements, specifications, UAT, training). Swiss resident, available immediately.

KEY SKILLS

AI & LLM

RAG · LangChain · OpenAI & Anthropic API · Prompt engineering · Embeddings · Vector stores · AI agents · Response quality evaluation

DATA ENGINEERING

Python · **SQL** · **PySpark** · **dbt** · ETL/ELT · Dimensional modeling (star, snowflake) · Data quality · CI/CD · Git

CLOUD & PLATFORMS

Databricks · **Microsoft Azure** · Azure Data Factory · Delta Lake · Snowflake · DuckDB

BI & ANALYTICS

Power BI (DAX) · SAP BusinessObjects (Web Intelligence) · Qlik · Advanced Excel · KPI design · Dashboards

DATABASES

Advanced SQL · Query optimisation · PostgreSQL · MongoDB · SAP BO semantic universes · Integrity controls

BUSINESS ANALYSIS & METHODS

Requirements gathering · Functional specifications · UAT & testing · Change management · Agile

TOOLS & ENVIRONMENTS

VS Code · Git & GitHub · GitHub Actions · Claude Code · Cursor · Copilot Studio · LangChain · Postman · Databricks Notebooks · Azure Data Factory · Power BI · SAP BO · Qlik · Talend · Jira · Confluence · Excel

PROFESSIONAL EXPERIENCE

EssilorLuxottica – Créteil (France) & Bangkok | Global group – 190,000 employees | Apprenticeship |
Oct. 2022 – Oct. 2025

Generative AI Engineer (LLM & RAG)

- Designed and deployed to production an **internal RAG chatbot** (LangChain, embeddings, vector store), cutting information search time by **30%** for the teams.
- Built the full **AI pipelines**: ingestion and cleaning of internal documentation sources, chunking, vector indexing and retrieval strategy.
- Integrated and **benchmarked several LLMs**; built sets of business questions to evaluate answer quality, relevance and consistency at each iteration.
- Iteratively optimised prompt engineering and retrieval strategies (prompt structure, search parameters, chunk size).
- Worked with business and IT teams to define use cases, measure impact and industrialise deployment; monitored answer quality after go-live.

Data Engineer

- Deployed and administered the **Databricks (Azure)** platform, processing multi-source data across **3 regions** (Europe, US, Asia).
- Automated **ETL pipelines in Python, SQL and PySpark**, ensuring reliable, high-performing data flows for decision-making.
- Orchestrated data flows with **Azure Data Factory** and **Delta Lake**, with versioning and integration through Git and CI/CD.
- Modeled data (**star and snowflake schemas**), built the SAP BO semantic universes and optimised queries and reports.
- Implemented **data quality and integrity controls** and technical documentation, easing maintenance and onboarding.

BI Business Analyst

- Contributed to the **BI migration from Qlik to SAP BusinessObjects**: gathered business requirements across Europe, wrote functional specifications and coordinated with technical teams.
- Took part in **user acceptance testing**: prepared test cases, ran validation sessions with business users and tracked defects through to resolution.
- Supported **change management**: training sessions delivered in French and English (Europe, US and Asia teams), illustrated documentation and post-migration support.
- Defined and formalised **business KPIs** (inventory, sales, supply chain) with management across Europe, US and Asia.
- Liaison between Business and IT teams across 3 continents for the inventory optimization project.

Data Analyst

- Designed and deployed **decision-making dashboards** (Power BI, SAP BO) tracking business KPIs for management across 3 continents.
- Ran ad hoc analyses for the global inventory optimization project, delivering actionable recommendations.
- Automated recurring reporting and defined upstream data quality rules, reducing anomalies in deliverables.

Airloop La Plage – Geneva | Leisure & hospitality | June 2026 – ongoing

Chef de rang (head waiter): service coordination and client relations in a high-traffic environment, alongside my search for an IT role.

Planzer – Geneva | Logistics & transport, Swiss market leader | 2026 – ongoing

Operational role in the supply chain: flow management in a demanding, high-tempo environment.

ENGINEERING ACADEMIC PROJECTS (CESI, 2021 – 2025)

- **AI applied to healthcare**: model predicting cancer recurrence risk from patient data: data preparation, anonymisation, training and comparison of several models, performance evaluation.
- **Hospital Big Data**: consolidation of heterogeneous sources for a hospital group: ETL chain, data warehouse, data cluster and statistical BI reporting.
- **Advanced algorithmics**: delivery route optimisation to reduce fuel consumption: operations research, complexity analysis and metaheuristics.
- **Cybersecurity**: post-incident analysis of a compromised information system: attack vectors, penetration testing and hardening recommendations.
- **Systems & networks**: integration of an acquired company's IS (Active Directory, virtualisation, containers, firewall) and compliance upgrade of a multi-country IS (VoIP, QoS, BGP/EIGRP).
- **Development**: automated file backup solution in client-server architecture (C#, design patterns, synchronisation, Git).

OPEN-SOURCE DATA & AI PROJECTS

Code, documentation and live demos: github.com/Toumil94

- **Swiss hotel tourism BI analysis** (github.com/Toumil94/analyse-bi-suisse): end-to-end BI project on official Swiss statistics (API, 400,000 rows): requirements, KPI dictionary, tested **star schema**, 4-page **Power BI** report with **24 DAX measures** stored as PBIP and validated in CI, UAT and an **executive memo with recommendations**.
- **RAG assistant for newcomers to Switzerland** (assistant-suisse.netlify.app): sourced answers on permits, health insurance or taxes from **official sources**: hybrid retrieval (BM25, query rewrites, RRF fusion), **free LLM** with extractive fallback, **40-question evaluation set** and automated tests.
- **Swiss rail punctuality (SBB/CFF)** (cff-ponctualite.netlify.app): **automated daily pipeline** (GitHub Actions) on Swiss transport open data: **DuckDB and dbt** with data tests, Parquet data lake and a **public dashboard** refreshed every morning.
- **Job Hunter** (github.com/Toumil94/job-hunter-ai): **AI agent** that collects Data and AI job offers in Switzerland, scores them against my profile and drafts tailored applications (Python, LLM, GitHub Actions).
- **Also**: **Market Data Pipeline** (Python, PostgreSQL, star schema, integrity controls) and **Professional CV Generator** (orchestration of several AI models, automated layout).

EDUCATION

Engineering degree in Computer Science, Data & AI specialisation — **2021 – 2025**
CESI Graduate School of Engineering – Lyon | CTI-accredited – MSc equivalent (Bac+5)
Project-based learning (A2P2): data, AI, databases, networks, cybersecurity and project management.

1st year BSc, Mathematics — **2019 – 2020**
Université Côte d'Azur Valrose – Nice, France

CERTIFICATIONS & LANGUAGES

- ✓ Azure Data Fundamentals (DP-900)
- ✓ Generative AI Fundamentals – Databricks
- ✓ Cisco CCNA – Networking
- ✓ Databricks Lakehouse Fundamentals
- ✓ AI Agent Fundamentals – Databricks
- ✓ TOEIC – 840/990 (B2)
- ✓ French – Native · English – Fluent (B2) · Spanish – Intermediate (B1)

REFERENCES

Mohamed Akram DJOUAHRA
Data Engineer – EssilorLuxottica

Thomas DELON
Project Manager – EssilorLuxottica

Amine CHABBI
Project Lead – EssilorLuxottica